

Performance Testing

Date	08 October 2026
Team ID	UNM1611BHAVANA
Project Name	AI FitTrack API
Maximum Marks	5 Marks

Team Members: BHAVANA, THEJASRI, TAMILVANA, VASIGARAN

Performance Testing

Performance testing evaluates how the AI FitTrack API behaves under expected and peak load conditions. The sections below document the testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (Collection Runner); optionally k6 / JMeter
Type of Testing	Load Testing and Stress Testing
Target Module / API	Login, Workout CRUD, Nutrition logging and AI recommendation endpoints
Test Environment	Local (Node.js + local MongoDB)
Test Date	08 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal load: login and fetch workouts	10	60	Avg response < 2 s, 0% errors
2	Create workout and meal log entries	25	60	Avg response < 2 s, error rate < 1%
3	AI recommendation requests	50	120	Max response < 5 s, error rate < 1%
4	Peak / stress load on all endpoints	100	120	System stays stable, error rate < 1%

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds			
2	Response Time (Max)	< 5 seconds			
3	Throughput (Req/sec)	>= 20 req/sec			
4	Error Rate	< 1%			
5	CPU Utilization	< 80%			
6	Memory Utilization	< 80%			

Evidence available: Login request = 782 ms; AI fitness-insights request = 17 ms. Throughput, CPU, and memory were not measured in the supplied screenshots.

Step 4: Observations & Analysis

Key Findings:

Simple CRUD endpoints are expected to respond well within the 2-second target on a local setup; the AI recommendation endpoints are the slowest because they process user history.

Bottlenecks Identified:

AI recommendation processing and unindexed MongoDB queries as data grows.

Optimization Steps Taken:

Add MongoDB indexes on `userId` and `date` fields, use pagination, return only required fields, and reuse the database connection.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., Postman runner results, k6 output):